Online Appendix to "A New Index of Financial Conditions"

Gary Koop
University of Strathclyde

Dimitris Korobilis*
University of Glasgow

August 14, 2014

*Corresponding author. Address: University of Glasgow, Adam Smith Business School, Gilbert Scott building, Glasgow, G12 8QQ, United Kingdom, Tel: +44 (0)141 330 2950, e-mail: Dimitris.Korobilis@glasgow.ac.uk

A. Data Appendix

The following table describes the series we used to extract our Financial Conditions Index. The series are selected as described in Section 3.1 of the paper. The fourth column describes the stationarity transformation codes (Tcodes) which have been applied to each variable. Tcode shows the stationarity transformation for each variable: Tcode=1, variable remains untransformed (levels) and Tcode=5, take first log differences. The fifth column describes the source of each variable. The codes are: B - Bloomberg; D - Datastream; F - Federal Reserve Economic Data (http://research.stlouisfed.org/fred2/); G - Amit Goyal (http://www.hec.unil.ch/agoyal/); R - Board of Governors of the Federal Reserve System (http://www.federalreserve.gov/); U - University of Michigan (http://www.sca.isr.umich.edu/); W - Mark W. Watson (http://www.princeton.edu/ mwatson/).

Table A1: The 18 variables used for the FCI

No	Mnemonic	Description	Tcode	Source	Sample
1	S&P500	S&P 500 Stock Price Index	5	F	1970:Q1-2013:Q3
2	TWEXMMTH	FRB Nominal Major Currencies Dollar Index (Linked To EXRUS In 1973:1)	5	W/F	1970:Q1-2013:Q3
3	OILPRICE	U.S. Crude Oil Imported Acquisition Cost by Refiners (Dollars per Barrel)	5	B	1974:Q1-2013:Q3
4	TED spread	3m LIBOR - 3m Treasury Bill Rate	1	B	1981:Q4-2013:Q3
5	10/2 y spread	10-Year/2-Year Treasury Yield Spread	1	F	1976:Q3-2013:Q3
6	2y/3m spread	2-Year/3-Month Treasury Yield Spread	1	F	1976:Q3-2013:Q3
7	Commercial Paper spread	3-Month Financial Commercial Paper/Treasury Bill Spread	1	B	1971:Q2-2013:Q3
8	LOANHPI Index	Home Loan Performance Index U.S. Index Level	5	B	1976:Q1-2013:Q3
9	30y Mortgage Spread	30y Conventional Mortgage Rate - 10y Treasury Rate	1	F	1971:Q2-2013:Q3
10	CMDEBT	Households and Nonprofit Organizations; Credit Market Instruments; Liability, Level	5	F	1971:Q1-2013:Q3
11	CCI Index	Continuous Commodity Futures Price Index	5	B	1970:Q1-2013:Q3
12	MOVE Index	Merrill Lynch One-Month Treasury Options Volatility Index (MOVE)	1	B	1988:Q2-2013:Q3
13	VIX	CBOE S&P100 Volatility Index	1	B	1986:Q3-2013:Q3
14	TOTALSL	Total Consumer Credit Owned And Securitized, Outstanding	5	F	1970:Q1-2013:Q3
15	STDSCOM	SLOOS: Net percentage of banks tightening standards for commercial real estate loans	1	R	1990:Q3-2013:Q3
16	MICH1	Michigan Survey: Interest Rates/Credit Reason for Good/Bad Conditions for Buying Houses Spread	1	U	1970:Q1-2013:Q3
17	MICH2	Michigan Survey: Buying Conditions for Vehicles	1	U	1970:Q1-2013:Q3
18	MICH3	Michigan Survey: Expected Change In Financial Situation	1	U	1970:Q1-2013:Q3

B. Technical Appendix

In this appendix, we describe the econometric methods we use to estimate a TVP-FAVAR and restricted versions of it.

We write the TVP-FAVAR compactly as

$$\begin{aligned} x_t &= z_t \Lambda_t + u_t, \quad u_t \sim N(0, V_t) && \text{(B.1)} \\ z_t &= z_{t-1}\beta_t + \varepsilon_t, \quad \varepsilon_t \sim N(0, Q_t) && \text{(B.2)} \\ \lambda_t &= \lambda_{t-1} + v_t, \quad v_t \sim N(0, W_t) && \text{(B.3)} \\ \beta_t &= \beta_{t-1} + \eta_t, \quad \eta_t \sim N(0, R_t) && \text{(B.4)} \end{aligned}$$

where $\lambda_t = \left((\lambda_t^y)', \left(\lambda_t^f\right)' \right)'$ and $z_t = \begin{bmatrix} y_t \\ f_t \end{bmatrix}$. We also use notation where $\widetilde{f}_t$ is the standard principal components estimate of f_t based on x_t (using data up to time t) and $\widetilde{z}_t = \begin{bmatrix} y_t \\ \widetilde{f}_t \end{bmatrix}$. Additionally, if a_t is a vector then $a_{i,t}$ is the i^{th} element of that vector; and if A_t is a matrix $A_{ii,t}$ is its $(i,i)^{th}$ element. Estimates of time varying parameters or latent states can be made using data available at time $t-1$ (filtering), or time t (updating) or time T (smoothing). We use subscript notation for this such that $a_{t|\tau}$ is an estimate (or posterior moment) of time-varying parameter a_t using data available through period τ.

Our estimation algorithm requires initialization of all state variables. In particular we define the following initial conditions for all system unknown parameters

$$\begin{aligned} f_0 &\sim N\left(0, \Sigma_{0|0}^f\right), && \text{(B.5)} \\ \lambda_0 &\sim N\left(0, \Sigma_{0|0}^\lambda\right), && \text{(B.6)} \\ \beta_0 &\sim N\left(0, \Sigma_{0|0}^\beta\right), && \text{(B.7)} \\ V_0 &\equiv 1 \times I_n, && \text{(B.8)} \\ Q_0 &\equiv 1 \times I_{s+1}. && \text{(B.9)} \end{aligned}$$

The algorithm follows the following steps:

1. Given the initial conditions and $z_t = \widetilde{z}_t$ obtain **filtered** estimates of $\lambda_t, \beta_t, V_t, Q_t$ using the following recursion for $t = 1, .., T$

   (a) The Kalman filter tells us:

$$\begin{aligned} \lambda_t | Data_{1:t-1} &\sim N\left(\lambda_{t|t-1}, \Sigma_{t|t-1}^\lambda\right), \\ \beta_t | Data_{1:t-1} &\sim N\left(\beta_{t|t-1}, \Sigma_{t|t-1}^\beta\right), \end{aligned}$$

where $\lambda_{t|t-1} = \lambda_{t-1|t-1}$, $\Sigma^{\lambda}_{t|t-1} = \Sigma^{\lambda}_{t-1|t-1} + \widehat{W}_t$, $\beta_{t|t-1} = \beta_{t-1|t-1}$ and $\Sigma^{\beta}_{t|t-1} = \Sigma^{\beta}_{t-1|t-1} + \widehat{R}_t$. The error covariances are estimated using forgetting factors as: $\widehat{W}_t = \left(1 - \kappa_3^{-1}\right) \Sigma^{\lambda}_{t-1|t-1}$ and $\widehat{R}_t = \left(1 - \kappa_4^{-1}\right) \Sigma^{\beta}_{t-1|t-1}$.

(b) Calculate estimates of V_t and Q_t for use in the updating step using the following EWMA specifications:

$$\widehat{V}_{i,t} = \kappa_1 V_{i,t-1|t-1} + (1-\kappa_1)\,\widehat{u}_{i,t}\widehat{u}'_{i,t} \tag{B.10}$$

$$\widehat{Q}_t = \kappa_2 Q_{t-1|t-1} + (1-\kappa_2)\,\widehat{\varepsilon}_t\widehat{\varepsilon}'_t \tag{B.11}$$

where $\widehat{u}_{i,t} = x_{i,t} - \widetilde{z}_t\lambda_{i,t|t-1}$, for $i = 1, ..., n$, and $\widehat{\varepsilon}_t = \widetilde{z}_t - \widetilde{z}_{t-1}\beta_{t|t-1}$.

(c) Update λ_t and β_t given information at time t using the Kalman filter update step

- Update $\lambda_{i,t}$ for each $i = 1, ..., n$ using

$$\lambda_{it}|Data_{1:t} \sim N\left(\lambda_{i,t|t}, \Sigma^{\lambda}_{ii,t|t}\right),$$

where $\lambda_{i,t|t} = \lambda_{i,t|t-1} + \Sigma^{\lambda}_{ii,t|t-1}\widetilde{z}'_t\left(\widehat{V}_{ii,t} + \widetilde{z}_t\Sigma^{\lambda}_{ii,t|t-1}\widetilde{z}'_t\right)^{-1}\left(x_t - \widetilde{z}_t\lambda_{t|t-1}\right)$ and $\Sigma^{\lambda}_{ii,t|t} = \Sigma^{\lambda}_{ii,t|t-1} - \Sigma^{\lambda}_{ii,t|t-1}\widetilde{z}'_t\left(\widehat{V}_{ii,t} + \widetilde{z}_t\Sigma^{\lambda}_{ii,t|t-1}\widetilde{z}'_t\right)^{-1}\widetilde{z}_t\Sigma^{\lambda}_{ii,t|t-1}$.

- Update β_t from

$$\beta_t|Data_{1:t} \sim N\left(\beta_{t|t}, \Sigma^{\beta}_{t|t}\right),$$

where $\beta_{t|t} = \beta_{t|t-1} + \Sigma^{\beta}_{t|t-1}\widetilde{z}'_{t-1}\left(\widehat{Q}_t + \widetilde{z}_{t-1}\Sigma^{\beta}_{t|t-1}\widetilde{z}'_{t-1}\right)^{-1}\left(\widetilde{z}_t - \widetilde{z}_{t-1}\widehat{\beta}_{t|t-1}\right)$ and $\Sigma^{\beta}_{t|t} = \Sigma^{\beta}_{t|t-1} - \Sigma^{\beta}_{t|t-1}\widetilde{z}'_{t-1}\left(\widehat{Q}_t + \widetilde{z}_{t-1}\Sigma^{\beta}_{t|t-1}\widetilde{z}'_{t-1}\right)^{-1}\widetilde{z}_{t-1}\Sigma^{\beta}_{t|t-1}$.

(d) Update V_t and Q_t given information at time t using the EWMA specifications as follows:

$$V_{i,t|t} = \kappa_1 V_{i,t-1|t-1} + (1-\kappa_1)\,\widehat{u}_{i,t|t}\widehat{u}'_{i,t|t} \tag{B.12}$$

$$Q_{t|t} = \kappa_2 Q_{t-1|t-1} + (1-\kappa_2)\,\widehat{\varepsilon}_{t|t}\widehat{\varepsilon}'_{t|t} \tag{B.13}$$

where $\widehat{u}_{i,t|t} = x_{i,t} - \widetilde{z}_t\lambda_{i,t|t}$, for $i = 1, ..., n$, and $\widehat{\varepsilon}_{t|t} = \widetilde{z}_t - \widetilde{z}_{t-1}\beta_{t|t}$.

2. Obtain **smoothed** estimates of $\lambda_t, \beta_t, V_t, Q_t$ using the following recursions for $t = T-1, .., 1$

(a) Update λ_t and β_t given information at time $t+1$ using the fixed interval smoother

- Update $\lambda_{i,t}$ for each $i = 1, ..., n$ from

$$\lambda_{it}|Data_{1:T} \sim N\left(\lambda_{i,t|T}, \Sigma^{\lambda}_{ii,t|T}\right),$$

where $\lambda_{i,t|T} = \lambda_{i,t|t} + C^{\lambda}_t\left(\lambda_{i,t+1|T} - \lambda_{i,t+1|t}\right)$, $\Sigma^{\lambda}_{ii,t|T} = \Sigma^{\lambda}_{ii,t|t} + C^{\lambda}_t\left(\Sigma^{\lambda}_{ii,t+1|T} - \Sigma^{\lambda}_{ii,t+1|t}\right)C^{\lambda\prime}_t$ and $C^{\lambda}_t = \Sigma^{\lambda}_{ii,t|t}\left(\Sigma^{\lambda}_{ii,t+1|t}\right)^{-1}$.

- Update β_t from

$$\beta_t|Data_{1:T} \sim N\left(\beta_{t|T}, \Sigma^{\beta}_{t|T}\right),$$

where $\beta_{t|T} = \beta_{t|t} + C^{\beta}_t\left(\beta_{t+1|T} - \beta_{t+1|t}\right)$, $\Sigma^{\beta}_{t|T} = \Sigma^{\beta}_{t|t} + C^{\beta}_t\left(\Sigma^{\beta}_{t+1|T} - \Sigma^{\beta}_{t+1|t}\right)C^{\beta\prime}_t$ and $C^{\beta}_t = \Sigma^{\beta}_{t|t}\left(\Sigma^{\beta}_{t+1|t}\right)^{-1}$.

(b) Update V_t and Q_t given information at time $t+1$ using the following equations

$$V^{-1}_{t|t+1} = \kappa_1 V^{-1}_{t|t} + (1-\kappa_1)\, V^{-1}_{t+1|t+1}, \qquad \text{(B.14)}$$

$$Q^{-1}_{t|t+1} = \kappa_2 Q^{-1}_{t|t} + (1-\kappa_2)\, Q^{-1}_{t+1|t+1}. \qquad \text{(B.15)}$$

3. Means and variances of f_t given appropriate estimates of $\lambda_t, \beta_t, V_t, Q_t$ described in the preceding steps can be obtained using the standard **Kalman filter and smoother.**

Treatment of missing values

In our application our sample is unbalanced, since it contains many financial variables which have been collected only after the 1970s or the 1980s. Similar issues are faced by organizations which monitor FCIs. For instance, the Chicago Fed National FCI comprises 100 series where most of them have different starting dates. Although specific computational methods for dealing with such issues exist (e.g. the EM algorithm or Gibbs sampler with data augmentation), our focus is on averaging over many models which means such methods are computationally infeasible. Accordingly, similar to our purpose of developing a simulation-free and fast algorithm for parameter estimation, we want to avoid simulation methods for estimating the missing data in x_t. Additionally, methods such as interpolation can work poorly when missing values are at the beginning of the sample.

Since the missing data in x_t are in the beginning, we make the assumption that the factor (FCI) is estimated using only the observed series. The estimation algorithm above allows for such an approach in a straightforward manner by just replacing missing values with zeros. The loadings λ (whether time-varying, or constant) will become equal to 0, thus removing from the estimate of f_t the effect of the variables in x_t which have missing values at time t. This feature holds both for the initial principal components estimate $\widetilde{f}_t$, as well as the final Kalman filter estimate.

Estimation of a single TVP-FAVAR

Given the algorithm above, we can estimate the TVP-FAVAR by choosing values of $\kappa_1, \kappa_2, \kappa_3, \kappa_4 < 1$. For the main results in the paper we set $\kappa_1 = \kappa_2 = 0.96$ and $\kappa_1 = \kappa_2 = 0.99$. The restricted special cases of the TVP-FAVAR listed in Section 2.1 can be obtained by setting forgetting and/or decay factors to particular values. If we set $\kappa_3 = 1$ and $\kappa_4 < 1$ then we can obtain a TVP-VAR augmented with factors estimated with constant loadings[1]. Setting $\kappa_3 = 1$ and $\kappa_4 = 1$ leads to the constant parameter FAVAR with heteroskedastic covariances (assuming that $\kappa_1, \kappa_2 < 1$). If we additionally set $\kappa_1 = \kappa_2 = 1$ we can estimate homoskedastic versions of the various models, since in that case $V_t = V_{t-1} = ... = V_1 = V_0$ and $Q_t = Q_{t-1} = ... = Q_1 = Q_0$. Nevertheless, as discussed in the main body of the paper, this is a case which is always dominated (in terms of forecast performance) by the heteroskedastic case. We provide more evidence for this in Appendix C.

Estimation of multiple models (DMA/DMS)

In order to implement the DMA/DMS exercises we run the algorithm described above for each of the $2^{17} = 131,072$ models. Note that for a specific DMA exercise all models are nested, and the only thing that changes is the number of variables in the vector x_t that we use in order to extract the FCI. Given our discussion about how missing values are treated by the Kalman filter, in order to estimate a specific model which uses, say, the 1st, 3rd and 15th series in x_t, we simply multiply all but the 1st, 3rd and 15th columns of x_t with zeros. In that case, we remove at all times t the effects of all 15 variables we do not use for estimation of the specific model, and at the same time we still have as a dependent variable a 18×1 dimensional vector (and programming is greatly simplified).

The most important feature of DMA is that, unlike many Bayesian model selection and averaging procedures which use MCMC methods, there is no dependence in estimating each model and iterations using "for" loops are independent. That means that it is trivial to adapt our code to use features such as parallel computing, thus taking advantage of the widespread availability of modern multi-core processors (or large clusters of PCs). In MATLAB this is as easy as replacing the typical "for" loop which would run for models 1 to $131,072$, with a "parfor" loop.

The reader is encouraged to look at our code which is available on https://sites.google.com/site/dimitriskorobilis/matlab, which also has the option to call the Parallel Processing Toolbox in a MATLAB environment.

[1] In a previous version of this paper we have named this model the factor augmented TVP-VAR or FA-TVP-VAR.

C. Sensitivity analysis

In this section we present further results for several different choices of prior hyperparameters, which can reflect different beliefs about time variation in the model parameters as well as beliefs about variation over time in the choice of the optimal model.

C.1. Comparison of relatively noninformative with training sample priors

In the main body of the paper results are presented for a subjectively-elicited but relatively noninformative prior. An interesting alternative is to choose all prior hyperparameters using a training sample of data. In the context of TVP-VARs, Primiceri (2005) suggests such a prior which is based on splitting the data into a training sample, and a testing sample where estimation occurs. OLS estimation of a constant coefficient model using the training sample provides parameter estimates which are used as prior hyperparameters for the testing sample. Such training sample priors are commonly-used in Bayesian analysis, and in the context of TVP models help provide regularized posterior estimators which can also help numerical stability. This latter feature is important in the case of TVP-VARs and TVP-FAVARs estimated with MCMC - see the discussion in Section 4.1 of Primiceri (2005).

In this section, we introduce such a training sample prior for our FAVAR and TVP-FAVAR models. We use the first 10 years of data (1970Q1-1979Q4) in our original sample as the training sample. We estimate a FAVAR with constant parameters using OLS methods (replacing the factor with its principal component estimate) on this training sample. These OLS estimates are used to in the initial conditions for the estimation sample, 1980Q1-2013Q3, as follows

$$
\begin{aligned}
f_0 &\sim N\left(\widehat{f}_T^{TS}, 0.1\right), \\
\lambda_0 &\sim N\left(0, 4 \times var\left(\widehat{\lambda}^{TS}\right)\right), \\
\beta_0 &\sim N\left(0, 4 \times var\left(\widehat{\beta}^{TS}\right)\right), \\
V_0 &\equiv 1 \times \widehat{V}^{TS}, \\
Q_0 &\equiv 1 \times \widehat{Q}^{TS}, \\
\pi_{0|0,j} &= \frac{1}{J},
\end{aligned}
$$

where parameters with a hat and a superscript TS denote OLS estimates of the respective parameters in the time-invariant FAVAR fitted using the training sample.

Other settings used in the main body of the paper remain the same, e.g. we use four lags everywhere and the decay and forgetting factors that define each of the models (DMA vs BMA, or FAVAR vs TVP-FAVAR) are exactly the ones specified in Section 3.2. The forecast evaluation sample is 1990Q1-2013Q3. Table C.1 presents the results of this exercise. It is divided in three blocks, each one corresponding to one of the three variables of interest (inflation, unemployment, output growth). For each block the first line shows the APL and MSFE of the benchmark TVP-FAVAR used in the main body of the paper. The APLs and MSFEs of all other models are relative to the APL and MSFE of the benchmark TVP-FAVAR, which implies that relative APL higher than one (similarly, relative MSFE lower than one) is an indication of better performance relative to the benchmark.

Table C1: Comparison of benchmark with training sample (TS) priors, 1990Q1-2013Q3

Forecast Metric	APL					MSFE				
					INFLATION					
	$h=0$	$h=1$	$h=2$	$h=3$	$h=4$	$h=0$	$h=1$	$h=2$	$h=3$	$h=4$
TVP-FAVAR	1.1233	0.9741	0.9066	0.8194	0.7380	0.0416	0.0454	0.0466	0.0562	0.0696
FAVAR	0.76	0.77	0.76	0.78	0.80	1.06	1.14	1.10	1.02	0.94
TVP-FAVAR-TS	1.01	1.04	1.06	1.10	1.09	0.93	0.94	0.87	0.87	0.97
FAVAR-TS	0.87	0.89	0.91	0.96	1.00	0.93	1.08	1.10	0.98	0.94
TVP-FAVAR-DMA-TS	1.13	1.13	1.15	1.16	1.17	0.94	0.84	0.89	0.90	0.86
TVP-FAVAR-BMA-TS	1.13	1.14	1.16	1.17	1.18	0.87	0.82	0.87	0.86	0.85
					UNEMPLOYMENT					
	$h=0$	$h=1$	$h=2$	$h=3$	$h=4$	$h=0$	$h=1$	$h=2$	$h=3$	$h=4$
TVP-FAVAR	0.9170	0.5868	0.4191	0.3229	0.2639	0.0906	0.2741	0.5950	1.0105	1.5359
FAVAR	0.64	0.61	0.61	0.63	0.64	1.68	2.01	2.09	2.14	2.17
TVP-FAVAR-TS	0.81	0.74	0.74	0.77	0.80	1.58	1.84	1.77	1.62	1.45
FAVAR-TS	0.50	0.51	0.56	0.62	0.66	2.44	2.85	2.82	2.47	2.21
TVP-FAVAR-DMA-TS	1.14	1.15	1.15	1.19	1.19	1.01	0.87	0.77	0.75	0.75
TVP-FAVAR-BMA-TS	1.15	1.16	1.16	1.19	1.22	1.00	0.86	0.77	0.75	0.75
					OUTPUT					
	$h=0$	$h=1$	$h=2$	$h=3$	$h=4$	$h=0$	$h=1$	$h=2$	$h=3$	$h=4$
TVP-FAVAR	0.4643	0.4344	0.4206	0.4156	0.4131	0.3548	0.4085	0.4204	0.3972	0.3854
FAVAR	0.71	0.71	0.72	0.73	0.74	1.43	1.64	1.57	1.48	1.40
TVP-FAVAR-TS	0.80	0.82	0.81	0.79	0.78	1.64	1.64	1.68	1.09	1.03
FAVAR-TS	0.72	0.76	0.74	0.66	0.65	2.04	1.84	2.01	1.21	1.37
TVP-FAVAR-DMA-TS	1.05	1.01	1.04	1.02	0.93	1.45	1.40	1.26	1.50	1.18
TVP-FAVAR-BMA-TS	0.97	0.95	1.06	0.93	0.89	1.44	1.40	1.26	1.50	1.18

Notes: APL is the average predictive likelihood (not in logarithms), and MSFE is the mean squared forecast error. Model's forecast performance is better when APL (MSFE) is higher (lower). For each variable (inflation, unemployment, output) the first line shows the APL and MSFE of the benchmark model for each forecast horizon h. All other models' APL and MSFE are relative to that of the benchmark model. Values of APL (MSFE) higher (lower) than 1 signify better performance than the benchmark.

In Table C1 we see that the training sample prior is not particularly helpful for forecasting unemployment and output, while there is marginal improvement of APLs and MSFEs for inflation. This result suggests that globally (for all three variables of interest) our relatively noninformative prior is a sensible choice and avoids problems with training samples such as that noted by Schorfheide and Wolpin (2012):

> "[...] from a Bayesian perspective the use of holdout samples is suboptimal because the computation of posterior probabilities should be based on the entire sample and not just on a subsample." Schorfheide and Wolpin (2012)

From our own experience with Bayesian VARs (and FAVARs), we can argue that training sample priors are very important in cases where numerical stability is an issue. For example, the decision of Primiceri (2005) and others to use training sample priors when estimating TVP-VAR with MCMC works well because serious numerical issues can occur when relatively noninformative priors and diffuse initial conditions are used in the full sample. In the present paper, where we examine high dimensional TVP-FAVARs, we do not have such numerical issues due to the computational simplicity of our algorithm (which does not involve the use of Monte Carlo or other iterative methods). This exact advantage of our estimation methods justifies our decision to use relatively noninformative priors in the full available sample as the benchmark case. In other datasets, e.g. macroeconomic data for the Euro-Area, training samples might not be available at all. In this case one can either use a subjectively elicited prior or a prior which is informed by economic theory. For example, economic theory restrictions can enter our prior distributions for the (FA)VAR part of our model, in the spirit of Filippeli and Theodoridis (2013) and Ingram and Whiteman (1994). Examining such restrictions is beyond the scope of our main aim in this paper, which is to show the general role of time-variation in models and parameters when extracting an FCI.

C.2. A Different Way to Purge Macroeconomic Conditions from the FCI

We have stressed in the main body of the paper that a good FCI should be purged of macroeconomic conditions. However, it might be argued that these macroeconomic conditions should not reflect the current situation, but rather current expectations of future macroeconomic conditions. In the body of the paper, we always use the current values of the three macroeconomic variables of interest (y_t). It is possible that y_t is a poor proxy of future macroeconomic expectations. In order to investigate this possibility, in this appendix we replace y_t by an explicit measure of macroeconomic expectations by using forecasts of the macroeconomic variables. In particular, we replace first equation of our TVP-VAR by

$$x_t = \lambda_t^y \widetilde{y}_t + \lambda_t^f f_t + u_t, \tag{C.1}$$

where $\widetilde{y}_t$ contains one-year ahead forecasts of inflation and output growth provided by the Survey of Professional Forecasters (SPF).[2] For the second model equation,

[2] This data is available at http://www.phil.frb.org/research-and-data/real-time-center/survey-of-professional-forecasters/historical-data/mean-forecasts.cfm

we maintain the assumption that the FCI f_t and the three macroeconomic variables of interest y_t follow a VAR of the form

$$\begin{bmatrix} y_t \\ f_t \end{bmatrix} = c_t + B_{t,1} \begin{bmatrix} y_{t-1} \\ f_{t-1} \end{bmatrix} + ... + B_{t,p} \begin{bmatrix} y_{t-p} \\ f_{t-p} \end{bmatrix} + \varepsilon_t. \tag{C.2}$$

In Table C2, we present the forecasting results for this model, which we name the TVP-FAVAR-SPF to denote the fact that we use the SPF in the measurement equation, and for the TVP-FAVAR and FAVAR as implemented in Section 3.4. Both TVP-FAVAR models have $\kappa_1 = \kappa_2 = 0.96$ and $\kappa_3 = \kappa_4 = 0.99$, but we additionally show results for the FAVAR-SPF which is the special case with $\kappa_1 = \kappa_2 = \kappa_3 = \kappa_4 = 1$ (a constant parameter FAVAR with the SPF variables used to purge the FCI). We can see that using the SPF variable in order to purge the FCI, results in a slight deterioration of forecast performance. It is only when using DMA with the TVP-FAVAR-SPFs, that we see improvement over our benchmark TVP-FAVAR, but DMA led to similar improvements in the body of the paper (i.e. it is the use of DMA which is causing these improvements, not the inclusion of the SPF variables).

Table C2: Comparison with the FAVAR with macroeconomic expectations (SPF), 1990Q1-2013Q3

Forecast Metric	APL					MSFE				
					INFLATION					
	$h=0$	$h=1$	$h=2$	$h=3$	$h=4$	$h=0$	$h=1$	$h=2$	$h=3$	$h=4$
TVP-FAVAR	1.1233	0.9741	0.9066	0.8194	0.7380	0.0416	0.0454	0.0466	0.0562	0.0696
FAVAR	0.76	0.77	0.76	0.78	0.80	1.06	1.14	1.10	1.02	0.94
TVP-FAVAR-SPF	1.01	0.99	1.00	1.00	0.99	1.01	0.99	1.01	0.98	1.03
FAVAR-SPF	1.00	0.98	0.99	1.01	1.02	1.16	1.22	1.23	1.20	1.15
TVP-FAVAR-DMA-SPF	1.12	1.12	1.13	1.15	1.16	1.05	1.04	1.10	1.09	1.05
TVP-FAVAR-BMA-SPF	1.14	1.08	1.15	1.15	1.12	1.16	1.05	1.11	1.20	1.18
					UNEMPLOYMENT					
	$h=0$	$h=1$	$h=2$	$h=3$	$h=4$	$h=0$	$h=1$	$h=2$	$h=3$	$h=4$
TVP-FAVAR	0.9170	0.5868	0.4191	0.3229	0.2639	0.0906	0.2741	0.5950	1.0105	1.5359
FAVAR	0.64	0.61	0.61	0.63	0.64	1.68	2.01	2.09	2.14	2.17
TVP-FAVAR-SPF	0.99	0.99	0.99	1.00	0.98	1.02	1.05	1.04	1.04	1.04
FAVAR-SPF	0.96	0.89	0.91	0.93	0.93	1.27	1.43	1.45	1.45	1.43
TVP-FAVAR-DMA-SPF	1.14	1.14	1.15	1.18	1.16	0.90	0.84	0.80	0.81	0.82
TVP-FAVAR-BMA-SPF	1.09	1.09	1.10	1.09	1.18	0.91	0.83	0.76	0.74	0.75
					OUTPUT					
	h=0	h=1	h=2	h=3	h=4	h=0	h=1	h=2	h=3	h=4
TVP-FAVAR	0.4643	0.4344	0.4206	0.4156	0.4131	0.3548	0.4085	0.4204	0.3972	0.3854
FAVAR	0.71	0.71	0.72	0.73	0.74	1.43	1.64	1.57	1.48	1.40
TVP-FAVAR-SPF	0.99	0.99	1.00	1.00	1.00	1.04	1.02	1.01	1.02	1.01
FAVAR-SPF	0.89	0.90	0.92	0.95	0.96	1.18	1.32	1.25	1.19	1.17
TVP-FAVAR-DMA-SPF	1.08	1.09	1.09	1.09	1.08	1.02	1.01	1.01	1.04	1.05
TVP-FAVAR-BMA-SPF	1.02	1.11	1.12	1.04	1.11	1.04	0.98	0.96	1.00	1.02

Notes: APL is the average predictive likelihood (not in logarithms), and MSFE is the mean squared forecast error. Model's forecast performance is better when APL (MSFE) is higher (lower). For each variable (inflation, unemployment, output) the first line shows the APL and MSFE of the benchmark model for each forecast horizon h. All other models' APL and MSFE are relative to that of the benchmark model. Values of APL (MSFE) higher (lower) than 1 signify better performance than the benchmark.

In general, from all the models we have attempted (also for different values of forgetting/decay factors or priors), we found that the benchmark specification consistently performs better than the specification in equations (C.1)-(C.2) which utilizes survey information.

C.3. Comparison of different rates model switching

Calculation of the DMA/DMS time-varying probabilities depends on selection of a hyperparameter α, which is a forgetting factor that determines how fast we "forget" past observations. Thus, this hyperparameter α controls how much data we use for the calculation of time-varying probabilities at time t and, thus, determines the rate of model switching. That is, as α gets lower only the most recent data are used and older data are discounted towards zero at a faster rate, resulting in model switching occurring at a faster rate.

We remind the reader that $\alpha = 1$ leads to standard BMA. The benchmark results reported in the body of the paper, using $\alpha = 0.99$, allow for slightly more rapid model switching. In this appendix we provide results for choices of α which reflect beliefs about even faster model switches, namely $\alpha = 0.96$, as well as the extreme case of $\alpha = 0.001$. Note that this latter case is close to the case of model averaging using equal weights, since it is trivial to prove that as $\alpha \rightarrow 0$ then the model weights degenerate to $\pi_{t|t-1,j} \rightarrow 1/J$ for all $j = 1, ..., J$. Naive model averaging schemes using equal weights have been shown in many cases to perform better than more elaborate econometric techniques that perform estimation of the model averaging weights; see Aiolfi, Capistrán and Timmermann (2010).

Table C3 shows the predictive likelihoods and mean squared forecast errors for the TVP-FAVAR model with DMA implemented for various values of forgetting factor α. As in the previous tables in this appendix, the first line shows the results for the benchmark TVP-FAVAR where all 18 variables are used to extract the FCI. All models have the benchmark prior described in the main text, plus the benchmark choices for decay factors of $\kappa_1 = \kappa_1 = 0.96$ and $\kappa_3 = \kappa_4 = 0.99$. We can immediately observe that the case $\alpha = 0.001$ results, in general, in much higher relative MSFEs for most horizons, for all three variables. The case $\alpha = 0.96$ sometimes does better than $\alpha = 0.99$ implying that there are periods in our sample that allowing for faster model switching would be beneficial. Therefore, there could potentially be further improvements in forecast accuracy by estimating α using grid-search methods, or even allowing a different value of α for each forecasting equation. We remind the reader that we do not perform such a search due to the already high computational demands of our empirical exercise, and we refer to Koop and Korobilis (2013) for an example of using such a procedure.

Table C3: Comparison of faster/slower model switching, 1990Q1-2013Q3

Forecast Metric	APL					MSFE				
					INFLATION					
	$h=0$	$h=1$	$h=2$	$h=3$	$h=4$	$h=0$	$h=1$	$h=2$	$h=3$	$h=4$
TVP-FAVAR	1.1233	0.9741	0.9066	0.8194	0.7380	0.0416	0.0454	0.0466	0.0562	0.0696
TVP-FAVAR-BMA ($\alpha=1$)	1.01	1.02	1.03	1.02	1.00	1.03	1.00	1.06	1.04	1.00
TVP-FAVAR-BMS ($\alpha=1$)	1.06	1.06	1.07	1.08	1.08	1.11	1.07	1.19	1.21	1.16
TVP-FAVAR-DMA ($\alpha=0.99$)	1.09	1.09	1.10	1.12	1.12	1.03	1.00	1.05	1.03	0.99
TVP-FAVAR-DMS ($\alpha=0.99$)	1.12	1.10	1.12	1.14	1.14	1.12	1.09	1.20	1.23	1.17
TVP-FAVAR-DMA ($\alpha=0.96$)	1.05	1.07	1.09	1.06	1.07	1.04	1.00	1.05	1.02	0.98
TVP-FAVAR-DMS ($\alpha=0.96$)	1.10	1.12	1.12	1.14	1.16	1.18	1.11	1.21	1.22	1.19
TVP-FAVAR-DMA ($\alpha=0.001$)	1.02	1.01	1.05	1.03	1.04	1.47	1.42	1.41	1.21	1.15
TVP-FAVAR-DMS ($\alpha=0.001$)	1.06	1.08	1.10	1.09	1.13	1.47	1.25	1.42	1.45	1.40
					UNEMPLOYMENT					
	$h=0$	$h=1$	$h=2$	$h=3$	$h=4$	$h=0$	$h=1$	$h=2$	$h=3$	$h=4$
TVP-FAVAR	0.9170	0.5868	0.4191	0.3229	0.2639	0.0906	0.2741	0.5950	1.0105	1.5359
TVP-FAVAR-BMA ($\alpha=1$)	1.01	1.01	0.97	0.96	0.90	0.91	0.85	0.82	0.83	0.84
TVP-FAVAR-BMS ($\alpha=1$)	1.07	1.05	1.04	1.03	1.00	0.97	0.88	0.83	0.83	0.84
TVP-FAVAR-DMA ($\alpha=0.99$)	1.11	1.10	1.09	1.10	1.07	0.92	0.86	0.83	0.84	0.85
TVP-FAVAR-DMS ($\alpha=0.99$)	1.08	1.09	1.09	1.11	1.10	0.99	0.86	0.79	0.80	0.81
TVP-FAVAR-DMA ($\alpha=0.96$)	1.05	1.09	1.06	1.08	1.10	0.89	0.82	0.80	0.80	0.82
TVP-FAVAR-DMS ($\alpha=0.96$)	1.13	1.12	1.13	1.19	1.14	0.95	0.84	0.76	0.74	0.75
TVP-FAVAR-DMA ($\alpha=0.001$)	0.99	1.00	1.01	0.95	1.02	1.08	0.89	0.82	0.81	0.83
TVP-FAVAR-DMS ($\alpha=0.001$)	1.08	1.04	1.07	1.12	1.03	1.03	0.88	0.77	0.76	0.76
					OUTPUT					
	$h=0$	$h=1$	$h=2$	$h=3$	$h=4$	$h=0$	$h=1$	$h=2$	$h=3$	$h=4$
TVP-FAVAR	0.4643	0.4344	0.4206	0.4156	0.4131	0.3548	0.4085	0.4204	0.3972	0.3854
TVP-FAVAR-BMA ($\alpha=1$)	0.95	0.94	0.93	0.91	0.91	0.97	0.95	0.94	0.97	0.98
TVP-FAVAR-BMS ($\alpha=1$)	1.00	1.00	0.99	0.99	0.99	1.01	0.99	0.98	1.01	1.00
TVP-FAVAR-DMA ($\alpha=0.99$)	1.03	1.04	1.03	1.03	1.04	0.96	0.94	0.94	0.97	0.98
TVP-FAVAR-DMS ($\alpha=0.99$)	1.05	1.06	1.06	1.08	1.09	0.99	0.95	0.95	1.00	1.00
TVP-FAVAR-DMA ($\alpha=0.96$)	1.03	1.04	1.04	1.02	1.01	1.01	1.00	1.00	1.03	1.04
TVP-FAVAR-DMS ($\alpha=0.96$)	1.10	1.07	1.08	1.12	1.11	1.04	0.99	0.97	1.01	1.03
TVP-FAVAR-DMA ($\alpha=0.001$)	0.96	0.98	0.96	0.96	0.94	1.06	1.03	1.04	1.05	1.10
TVP-FAVAR-DMS ($\alpha=0.001$)	1.08	0.97	0.99	1.06	1.04	1.08	1.03	1.00	1.03	1.05

Notes: APL is the average predictive likelihood (not in logarithms), and MSFE is the mean squared forecast error. Model's forecast performance is better when APL (MSFE) is higher (lower). For each variable (inflation, unemployment, output) the first line shows the APL and MSFE of the benchmark model for each forecast horizon h. All other models' APL and MSFE are relative to that benchmark model. Values of APL (MSFE) higher (lower) than 1 signify better performance than the benchmark.

C.4. Comparison of different rates of parameter change

Similar to the hyperparameters that control model switching, $\kappa_1, \kappa_2, \kappa_3, \kappa_4$ control the amount of time-variation in the error covariances (V_t, Q_t), as well as the time-varying loadings λ_t and the VAR coefficients β_t. Table C4 presents results for different choices of these decay and forgetting factors. We only present results for the TVP-FAVAR models with all 18 variables used to extract the FCI, and not for the more computationally intensive DMA/DMS variants of the TVP-FAVAR (which allow selection of the optimal number of variables to include in the FCI). The first line again shows the benchmark TVP-FAVAR with $\kappa_1 = \kappa_2 = 0.96$ and $\kappa_3 = \kappa_4 = 0.99$. As for the other tables in this appendix, we present relative APLs and relative MSFEs, for the evaluation period 1990Q1-2013Q3.

Table C4: Comparison of faster/slower parameter switching, 1990Q1-2013Q3

Forecast Metric			APL					MSFE		
					INFLATION					
	$h=0$	$h=1$	$h=2$	$h=3$	$h=4$	$h=0$	$h=1$	$h=2$	$h=3$	$h=4$
FAVAR (1,1)	0.8584	0.7474	0.6924	0.6427	0.5934	0.0438	0.0524	0.0542	0.0604	0.0686
TVP-FAVAR (0.96,1)	1.36	1.32	1.34	1.30	1.30	1.12	1.05	1.01	0.99	1.00
TVP-FAVAR (0.94,1)	1.49	1.45	1.49	1.47	1.46	1.12	1.02	0.98	0.94	0.98
TVP-FAVAR (0.92,1)	1.53	1.51	1.52	1.52	1.50	1.11	1.02	0.99	0.97	1.00
TVP-FAVAR (1,0.99)	1.01	1.01	1.02	1.00	1.00	0.97	0.96	0.97	0.98	1.02
TVP-FAVAR (0.96,0.99)	1.34	1.34	1.35	1.32	1.29	0.95	0.86	0.86	0.93	1.01
TVP-FAVAR (0.94,0.99)	1.46	1.44	1.48	1.45	1.43	1.05	0.95	0.92	0.88	0.95
TVP-FAVAR (0.92,0.99)	1.50	1.47	1.49	1.49	1.45	1.08	0.99	0.96	0.94	0.99
					UNEMPLOYMENT					
	$h=0$	$h=1$	$h=2$	$h=3$	$h=4$	$h=0$	$h=1$	$h=2$	$h=3$	$h=4$
FAVAR (1,1)	0.5914	0.3567	0.2557	0.2024	0.1676	0.1200	0.4102	0.9406	1.6578	2.5544
TVP-FAVAR (0.96,1)	1.65	1.77	1.83	1.87	1.88	0.70	0.59	0.56	0.56	0.56
TVP-FAVAR (0.94,1)	1.74	1.90	1.95	2.00	1.99	0.71	0.60	0.56	0.55	0.55
TVP-FAVAR (0.92,1)	1.74	1.89	1.96	2.01	2.03	0.73	0.62	0.58	0.56	0.56
TVP-FAVAR (1,0.99)	1.09	1.13	1.18	1.21	1.23	1.03	1.01	0.99	0.97	0.97
TVP-FAVAR (0.96,0.99)	1.60	1.73	1.76	1.74	1.75	0.69	0.59	0.55	0.54	0.54
TVP-FAVAR (0.94,0.99)	1.73	1.85	1.88	1.90	1.87	0.65	0.53	0.51	0.51	0.52
TVP-FAVAR (0.92,0.99)	1.77	1.94	1.95	1.98	1.95	0.65	0.53	0.50	0.50	0.51
					OUTPUT					
	$h=0$	$h=1$	$h=2$	$h=3$	$h=4$	$h=0$	$h=1$	$h=2$	$h=3$	$h=4$
FAVAR (1,1)	0.3276	0.3071	0.3031	0.3016	0.3041	0.3694	0.5209	0.5542	0.5181	0.4967
TVP-FAVAR (0.96,1)	1.54	1.56	1.51	1.53	1.50	0.91	0.78	0.77	0.81	0.84
TVP-FAVAR (0.94,1)	1.63	1.63	1.62	1.62	1.62	0.90	0.78	0.76	0.80	0.84
TVP-FAVAR (0.92,1)	1.66	1.68	1.66	1.68	1.68	0.92	0.77	0.76	0.80	0.83
TVP-FAVAR (1,0.99)	1.14	1.14	1.13	1.12	1.11	1.13	1.01	0.95	0.92	0.91
TVP-FAVAR (0.96,0.99)	1.51	1.51	1.49	1.48	1.46	0.91	0.75	0.72	0.75	0.77
TVP-FAVAR (0.94,0.99)	1.59	1.58	1.57	1.55	1.55	0.91	0.76	0.74	0.79	0.83
TVP-FAVAR (0.92,0.99)	1.62	1.64	1.62	1.60	1.61	0.91	0.75	0.74	0.78	0.82

Notes: APL is the average predictive likelihood, and MSFE is the mean squared forecast error. Model's forecast performance is better when APL (MSFE) is higher (lower). For each variable (inflation, unemployment, output) the first line shows the APL and MSFE of the benchmark model for each forecast horizon h. All other models' APL and MSFE are relative to that of the benchmark model. Values of APL (MSFE) higher (lower) than 1 signify better performance than the benchmark.

It is interesting to note that our conservative benchmark prior on time-variation in the volatilities ($\kappa_1 = \kappa_2 = 0.96$) is globally dominated by a prior which allows faster switching in volatilities ($\kappa_1 = \kappa_2 = 0.92$). This result is not surprising, as there is ample evidence for a high degree of volatility in macroeconomic and financial data (usually modeled in the literature as a geometric random walk, or a persistent AR(1) process). It would be surprising to find support for large time-variation in the time-varying parameters λ_t,β_t. However, we do not find such support. In terms of predictive likelihoods results are similar for $\kappa_3 = \kappa_4 = 1$ (constant parameters) and $\kappa_3 = \kappa_4 = 0.99$, while MSFEs seem to favor a small degree of time-variation. However, values of κ_3 and κ_4 of 0.98 or lower, forecasting results deteriorate dramatically, showing the consequences of allowing too large a degree of time variation in the VAR coefficients and factor loadings.

References

[1] Aiolfi, M., Capistrán, C., Timmermann, A., 2010. Forecast combinations. In: Michael P. Clements and David F. Hendry (Eds). The Oxford Handbook of Economic Forecasting. Oxford University Press, USA.

[2] Filippeli, T., Theodoridis, K., 2013. DSGE priors for BVAR models, Empirical Economics, forthcoming. doi: 10.1007/s00181-013-0797-z

[3] Ingram, B.,Whiteman,C.,1994.Supplanting the 'Minnesota' prior: forecasting macroeconomic time series using real business cycle model priors. Journal of Monetary Economics 34, 497-510.

[4] Schorfheide, F., Wolpin, K. E., 2012. On the use of holdout samples for model selection. American Economic Review - Papers and Proceedings 102(3), 477-481